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Tackle characteristics associated with concussion in elite men's rugby union: unpicking the differences between tacklers and ball-carriers

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ABSTRACT

Objective To identify characteristics of tackling, of being tackled and interactions between tackle characteristics that are associated with concussion.

Methods A case-control study in male professional rugby union players in England over five seasons (2016/2017 to 2020/2021) analysed characteristics of tackles that led to a clinically diagnosed concussion (cases), and a control group of tackles that did not result in a concussion. ORs were plotted against the overall frequency of each tackle characteristic.

Results 231 tackles resulting in concussions (tackler 178, 77%; ball-carrier 53, 23%), alongside 9963 control tackles, were analysed. For tacklers, 'head to torso' ($_{\text{Lower CI}}$ OR $_{\text{Upper CI; 4.0}}$ 6.5 $_{10.7}$) had relatively low odds of concussion compared with 'no head contact'. 'Head to knee' had the highest odds of concussion ($_{75.3}$ 155.8 $_{322.4}$), but 'head to hip' occurred more frequently and had the highest number of concussions per 1000 tackles (3.1/1000 tackles). For ball-carriers, 'head to head' contact had the highest odds of concussion ($_{56.7}$ 104.3 $_{192.0}$). When 'tackler body position' was 'upright', the odds of concussion to the tackler were significantly higher when contacting the ball-carrier's 'head and neck' versus their 'torso' ($_{3.0}$ 23.7 $_{206.7}$).

Conclusions Lower tackles reduce the chances of concussion to ball-carriers. The influence of tackle height on concussion to tacklers is more nuanced, but the chances are relatively low when contact is made with the ball-carrier's torso. These findings support ongoing implementation of strategies to reduce concussion risk by lowering tackle height to target the torso.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ The characteristics of tackles that have a higher chance of resulting in concussion to one of the players involved have been described in rugby union. The ball-carrier and tackler(s) concussion risk has not previously been considered separately. The characteristics of tackles resulting in any injury have also been described and these differ for the ball-carrier and the tackler.

WHAT THIS STUDY ADDS

⇒ This study deepens our understanding of the characteristics of tackles that have a higher chance of causing concussion. Specifically, it demonstrates that the odds of a concussion to the ball-carrier are lower when the point of contact on the ball-carrier is lower. The odds of concussion to the tackler are more nuanced, but when contact is made with the torso, the odds of concussion are relatively low.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings contribute to the evidence base informing strategies to prevent concussion in rugby union, including current discussions around lowering the maximum legal height of the tackle in elite rugby union.

INTRODUCTION

In elite men's rugby union in England, match concussion incidence has been reported to be 20 per 1000 hours (95% CI 18 to 24), representing around 20% of all match injuries.[1] Between 50%[2] and 74%[3] of concussions are associated with the tackle event, making it a key focus for concussion prevention efforts.

The characteristics of tackles resulting in off-field head injury assessment (HIA)[4] and physician diagnosed concussion[5] were previously examined in elite men's rugby union. Tackles that were active shoulder, front-on, high speed or involved an accelerating tackler had a higher propensity to result in an HIA.[4] Tackles where the tackler, or both the tackler and ball-carrier, were accelerating, where the tackler was moving at high speed, where there was head contact and high tackles were more likely to result in a concussion for one of the players involved.[5]

Studies in elite rugby union have not previously examined the characteristics of tackles resulting in concussion specifically to the tackler or the ball-carrier. Different

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characteristics may be associated with concussion according to a player's role (ie, tackler or ball-carrier). For all injuries, lower tackles are more likely to result in injury to the tackler, and higher tackles are more likely to result in injury to the ball-carrier.[6] In elite men's rugby league, entering the tackle in an upright position had a greater propensity of HIA[7] and concussion[8] for tacklers and for ball-carriers. A further consideration is that the complexity of the tackle event lends itself to exploring the interaction of different tackle characteristics.

The aims of this study were to identify (1) characteristics that are associated with concussion specifically to players when tackling and when being tackled, and (2) interactions between characteristics that are associated with concussion in the tackle event.

Methods

Participants

This case-control study was conducted in male professional rugby union players competing in The English Premiership over five seasons (2016/2017 to 2020/2021). Written informed consent was obtained from each participant as part of the Professional Rugby Injury Surveillance Project.

Procedures

This exploratory study broadly replicates the procedures of Cross *et al*.[5] The tackle was defined as 'any event where one or more tacklers attempted to stop or impede the carrier and the ball-carrier was brought to ground'.[9]

Over the period of data collection, the HIA process was in place to aid clinical decision-making (in-game) and subsequent diagnosis (postgame) of concussion. This multipoint in-time assessment process and the operational definition of concussion in professional rugby union have been explained previously[10] and formed the basis for the physician-led diagnosis of concussion. Concussion cases were prospectively reported as part of the Professional Rugby Injury Surveillance Project, which aligned with relevant consensus statements.[11 12] Club medical staff reported detailed injury information which was used to identify the specific match event that led to the player being diagnosed with concussion. Any concussions reported as delayed or evolving were not included because they could not be directly associated with a specific match event.

Characteristics of tackles that led to a clinically diagnosed concussion (cases), and control tackles coded from the same fixtures that did not result in a concussion (25 tackles immediately preceding and proceeding the case event), were collated from available match coding data (Opta, Stats Perform, London, UK). Concussions sustained near the beginning or end of a match may not have had 25 control events either side of the case event (in which cases as many events that occurred were included), and if two players sustained a concussion in the same tackle event, control tackles were not duplicated.

A single experienced video analyst coded tackle events using a predefined coding matrix on a bespoke analysis platform (Fair Play, Australia). The coding matrix comprised 20 categorical variables (see online supplemental A) developed from previously used templates.[5 13] Video clips were excluded if footage quality precluded (1) the match event to be clearly observed or (2) the application of the coding template to the tackle.

Data analysis

All estimations were performed using R (V.4.3.2, R Foundation for Statistical Computing, Vienna, Austria). Due to the large number of potential variables, a variable selection process was undertaken to identify the tackle characteristics most highly related to the response variable (ie, concussion) for interpretation.[14] The random forest algorithm from the *randomForest* package[15] was trained with the number of trees (*ntree*) set to 500, and default settings for other hyperparameters (eg, *mtry*=square root of the number of predictors, *nodesize*=1). Variable importance was determined using the 'mean decrease accuracy' metric, with key tackle characteristics selected based on visual inspection of the importance plot combined with domain expertise. Sensitivity analyses using hyperparameter tuning (eg, varying *mtry* and *nodesize*) and k-fold cross-validation were conducted but did not alter the selected variables. Thus, the simpler model with default hyperparameters was retained. To ensure robust model evaluation and address potential data leakage, a player-stratified train-test (70:30) split was implemented to prevent the same player's data from appearing in both training and test sets. Model performances were evaluated based on the area under the receiver operating characteristic curve (AUROC) using a hold-out dataset comprising 30% of the data;[16] <0.6, *fail*; 0.6≤AUROC<0.7, *poor*; 0.7≤AUROC<0.8, *fair*; 0.8≤AUROC<0.9, *good*; ≥0.9, *excellent*.

Subsequently, a univariate generalised linear mixed model (family=binomial) was implemented via the *lme4* package[17] to model the association between retained variables and risk of concussion. Tackle characteristics were placed in the model as fixed effects, season was included as a (categorical) covariate and player ID was included as a random effect to account for clustering of observations. The tackle characteristic level with the highest frequency was set as the reference category. ORs were calculated by exponentiating the log-odds coefficients from the fixed effects, and 90% CIs were derived by exponentiating the Wald-based CIs of the log-odds. Separate analyses were run for concussions sustained by the ball-carrier and tackler, alongside a combined analysis to enable comparisons to past analyses. Given the previous identification of tackle height as a risk factor for concussion, the odds of concussion to the tackler and the ball carrier were also explored according to the first point of contact on the ball carrier. A selection of interaction effects was defined a priori to be explored, based on previous literature and domain expertise. These were; body position interactions

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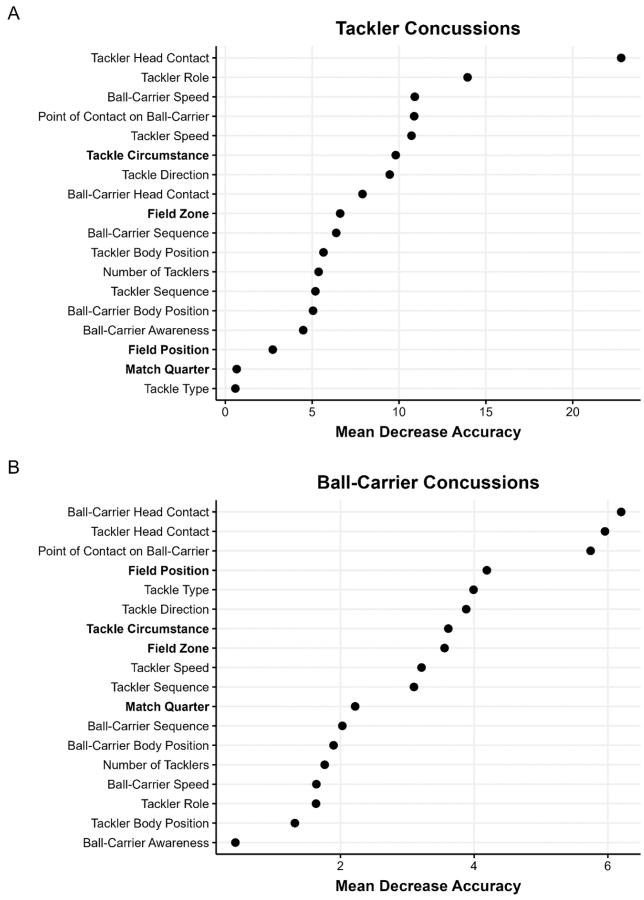

Figure 1 Variable importance plots for Tackler concussions (A) and Ball-Carrier concussions (B) displaying the decrease in model accuracy for classifying concussion (change in out-of-bag (OOB) error) when each variable is excluded. The higher the mean decrease accuracy (MDA) value, the more important the variable is for successful classification. Variables in bold are from Opta data, with the remaining variables derived from bespoke tackle analysis.

(ie, tackler body position×ball-carrier body position), tackle speed interactions (eg, ball-carrier speed×tackler speed and ball-carrier speed×tackler body position) and factors that have been the focus of concussion reduction efforts (eg, tackler body position×point of contact on ball-carrier). The interaction effects were explored separately for the tackler and ball-carrier concussions. The *emmeans* package[18] was used to explore pairwise contrasts within each level of interaction on the response scale, with ORs and 90% CIs calculated by comparing predicted probabilities against the reference category (highest frequency level).

ORs were plotted against the overall frequency of each tackle characteristic to allow interpretation of both the risk associated with the given tackle characteristic (in the current data set) and the regularity with which it occurred.[19] Minimal effects tests were applied to the ORs, using a smallest effect size of interest (SESOI) (OR) of 0.67 and 1.50,[20] and declared as significant when the 90% CI fell beyond the SESOI.[21] The OR and 90% CIs are reported as: $_{\text{Lower CI}}\,\text{OR}_{\text{Upper CI}}$.[22]

RESULTS

In total, of 425 reported tackle concussions over the study period, 231 concussion events (54%) could be definitively identified on video. Of reported concussions, 290 (68%) occurred to the tackler and 135 (32%) to the ball-carrier. Of the video identified concussion events, 178 (77%) occurred to the tackler and 53 (23%) to the ball-carrier. The control sample consisted of 9963 tackles. The variable selection process identified 'tackler head contact' as the most important variable for all concussions and concussions sustained by the tackler. For ball-carrier concussions, 'ball-carrier head contact', 'tackler head contact' and 'point of contact on ball-carrier' were identified as key variables for further analysis (figure 1). The AUROC values for the overall, tackler and ball-carrier random forest models were 0.76, 0.84 and 0.73, respectively.

For tackler concussions, all head contact locations significantly increased the odds of concussion compared with 'no head contact'. 'Head to knee' was associated with the highest odds of concussion ($_{75.3}\,155.8\,_{322.4}$), followed by 'head to teammate head' ($_{44.7}\,78.0\,_{136.1}$), 'head to hip' ($_{30.0}\,45.7\,_{69.4}$) and 'head to head' ($_{18.7}\,35.3\,_{66.7}$). Of these, 'head to hip' occurred more frequently than 'head to knee' and so was associated with the highest number of concussions per 1000 tackles. 'Head to torso' ($_{4.0}\,6.5\,_{10.7}$), 'head to arm' ($_{4.6}\,7.8\,_{13.1}$) and 'head to upper leg' ($_{5.4}\,8.9\,_{14.7}$) were associated with the lowest odds of concussion (figure 2A).

For ball-carrier concussions, the 'ball-carrier head contact' variable identified 'head to head' as carrying the highest risk of concussion ($_{56.7}\,104.3\,_{192.0}$) versus 'no head contact', followed by 'head to shoulder' ($_{7.2}\,15.0\,_{31.1}$) and 'head to arm' ($_{2.5}\,5.2\,_{10.7}$). All other contact locations were inconclusive (figure 2B). For the 'tackler head contact' variable, 'head to head' contact was associated with the highest risk of concussion ($_{39.5}\,70.4\,_{125.4}$), followed by 'whiplash injury' ($_{5.0}\,29.8\,_{178.7}$) and 'head to shoulder' contact ($_{3.3}\,7.4\,_{16.6}$); all other 'tackler head contact' locations were inconclusive (figure 2C). For 'point of contact on ball-carrier', the 'head and neck' was associated with a significantly higher odds of concussion ($_{15.2}\,27.2\,_{48.7}$) versus the reference category (torso); all other locations were inconclusive (figure 2D).

For the combined tackler and ball-carrier analysis, 'head to knee' was associated with the highest odds of concussion ($_{41.1}\,80.1\,_{156.1}$), followed by 'head to head' ($_{33.2}\,51.3\,_{79.3}$) and 'head to teammate head' ($_{27.0}\,44.6\,_{73.5}$). 'Head to torso' ($_{2.5}\,3.8\,_{5.9}$), 'head to arm' ($_{3.0}\,4.8\,_{7.5}$) and 'head to upper leg' ($_{3.7}\,5.7\,_{8.8}$) were associated with the lowest odds of concussion (online supplemental B).

46% of tackles made first point of contact to the ball-carrier's torso (figure 3). For the tackler, the odds of concussion when first point of contact was to the ball-carrier's upper leg was $_{1.6}\,2.3\,_{3.2}$ versus the torso, with 20%

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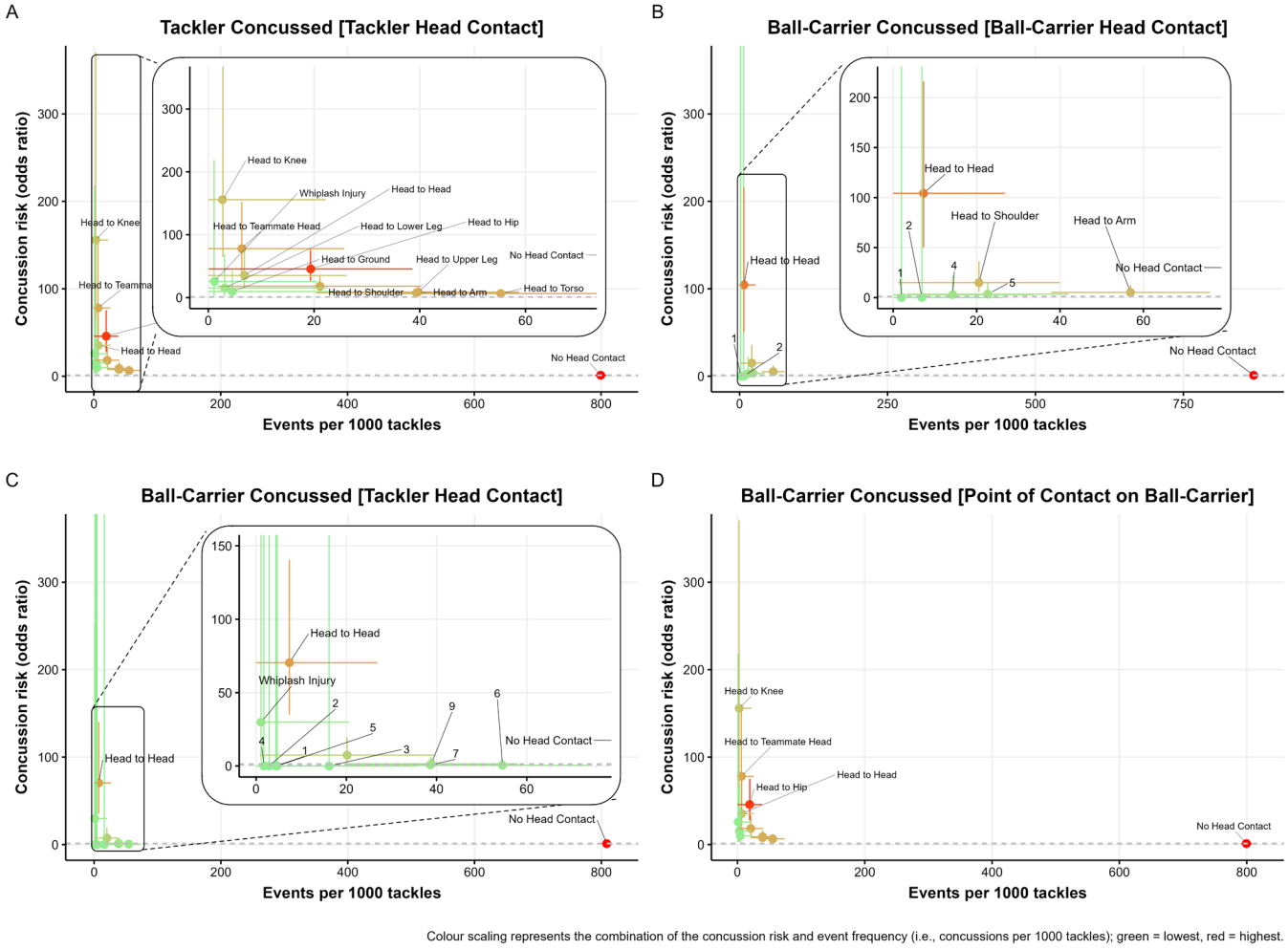

Figure 2 Likelihood of concussion for each variable comparing characteristics with the reference (most common) category (y-axis) by the overall frequency of each characteristic (x-axis) for (A) tackler concussion, tackler head contact, (B) ball-carrier concussion, ball-carrier head contact (1=Whiplash injury, 2=Head to upper leg, 4=Head to torso, 5=Head to ground), (C) ball-carrier concussion, tackler head contact (1=Head to ground, 2=Head to lower leg, 3=Head to hip, 4=Head to knee, 5=Head to teammate head, 6=Head to torso, 7=Head to arm, 9=Head to upper leg) and (D) ball-carrier concussion, point of contact on ball-carrier (1=Lower leg, 2=Upper leg, 4=Shoulder and armpit). Characteristics significantly different from the reference category are labelled; characteristics with a number label were not significantly different from the reference category.

of tackles making contact in this area. For the ball-carrier, the odds of concussion when first point of contact was to the ball-carrier's head and neck was $_{13.6}\,27.2\,_{54.5}$ versus the torso, with 2% of tackles making contact in this area.

From the interaction effects explored for tackler concussions, only 'ball-carrier speed×tackler speed' and 'point of contact on ball-carrier×tackler body position' were statistically significant. For 'ball-carrier speed×tackler speed', when 'ball-carrier speed' was 'static', the odds of concussion were significantly increased when the tackler was moving at 'high speed' versus 'static' ($_{5.3}\,1088.3\,_{222222.2}$) or 'in motion' ($_{2.5}\,11.9\,_{57.9}$). When 'ball-carrier speed' was 'in motion', the odds of concussion were significantly increased when the tackler was moving at 'high speed' versus 'static' ($_{2.3}\,7.3\,_{22.6}$) or 'in motion' ($_{2.1}\,6.6\,_{20.9}$). When ball-carrier speed was 'high speed', the effects of 'tackler speed' were inconclusive (figure 4A; online supplemental C and D).

For 'point of contact on ball-carrier×tackler body position', when 'tackler body position' was 'upright', the odds of concussion were significantly higher when 'point of contact on ball-carrier' was 'head and neck' versus 'torso' ($_{3.0}\,23.7\,_{206.7}$). When 'tackler body position' was 'bent at waist', the odds of concussion were significantly higher when 'point of contact on ball-carrier' was 'upper leg' versus 'shoulder and armpit' ($_{1.7}\,3.2\,_{6.2}$). All other comparisons were inconclusive (figure 4B).

DISCUSSION

This study examined tackle characteristics that influence concussion risk in elite men's rugby union. A point to note is that, while there are positional variations in the number and the type of tackles that players make, all players are tacklers and ball-carriers in a match. Understanding the characteristics of tackler and ball-carrier behaviours that are more likely to result in concussion

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% of tackles	Tackler	Ball-Carrier
Head & Neck 2%	1.7 (0.7-4.3)	**27.2 (13.6-54.5)**
Shoulder & Armpit 30%	0.8 (0.5-1.2)	1.6 (0.8-3.3)
Torso 46%	Reference	Reference
Upper Leg 20%	**2.3 (1.6-3.2)**	0.1 (<0.1-1.0)
Lower Leg 2%	1.0 (0.3-3.1)	<0.1 (<0.1-313.6)

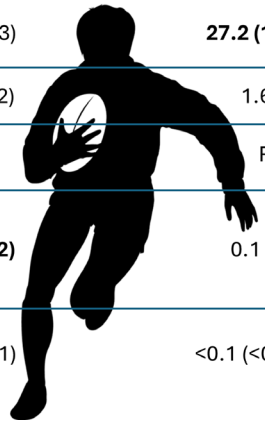

Figure 3 Likelihood of concussion to tacklers and ball-carriers according to the first point of contact on the ball-carrier. See online supplemental A for location definitions. Bold text highlights significant difference from the reference category.

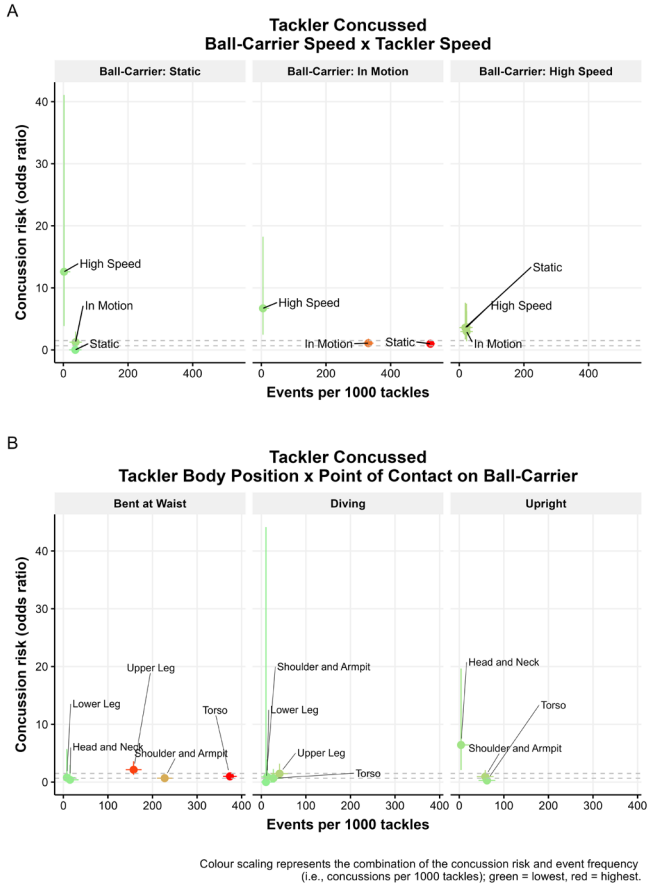

Figure 4 Likelihood of concussion for interactions between variables comparing characteristics with the reference (most common) category (y-axis) by the overall frequency of each characteristic (x-axis) for (A) tackler concussion, ball-carrier speed × tackler speed, (B) tackler concussion, tackler body position × point of contact on ball-carrier.

can inform strategies for concussion reduction for all players. For concussions to tacklers, 'head to knee', 'head to hip' and 'head to teammate head' had the highest odds of concussion, whereas 'head to head' contact had the highest odds for the ball-carrier. When considering interactions in the tackle, concussions sustained to the tackler had higher odds when the tackler was moving at high speed; in addition, when the tackler was upright and the first point of contact was to the ball-carrier's head or neck, the odds of concussion were high. Player speed and direction are interesting characteristics to explore for context, but body position and height of contact in the tackle are characteristics that are much more amenable to change.

When combining ball-carrier and tackler, 'head to head' contact resulted in elevated odds of concussion, consistent with previous analysis.[5] It is notable that 'head to head' is the only head contact type that featured in all of the ball-carrier concussed and tackler concussed analyses. 'Head to knee' was also associated with high odds of concussion in the combined analysis. Most (77%) of the concussions analysed in this study occurred to the tackler, whereas 68% of reported concussions to the tackler over the study period is similar to what has previously been reported using injury surveillance unsupported by systematic video analysis of the concussion event, whereby 54–67% of concussions in the tackle were reported for the tackler.[23] There may be an under-representation of ball-carrier concussion in the current study due to the challenge of identifying some ball-carrier concussion events. Specifically, this was the case with so-called 'pick and go' scenarios, when a ball-carrier picked the ball up from the ground and moved at relatively low speed while maintaining a very low body position when contacting the tackler, and when players carried the ball into multiple tackles in succession.

For tackler concussions, 'head to knee', 'head to hip' and 'head to head of a teammate' were head contact types that had the highest odds of concussion, but a higher frequency of 'head to hip' meant that this resulted in more concussions than the other locations. In contrast, 'head to torso' and 'head to upper leg' had much lower odds of concussion. Tackle coaching to optimise safe and effective tackles focuses on making initial contact with the ball-carrier's torso;[24] with current findings indicating that there should be a coaching focus on avoiding the tackler's head making contact with the ball-carrier's hip and a team mate's head. In addition, existing focus on sanctioning head contact in elite rugby union[25] and law changes lowering the permitted height of the tackled in community rugby union both aim to increase tackles to the torso. These findings support continued efforts to increase tackles to the ball-carrier's torso.

Very low tackles resulting in the tackler's head making contact with the ball-carrier's knee had high odds of concussion. One criticism of law changes lowering the permitted height of tackles in community rugby union has been that it is likely to increase head to knee contact.

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Ball carriers may lower their body position specifically to make it more challenging to make an effective tackle, and although the goal is not to reduce tackle height to target the knees, tacklers may not have any choice but to tackle very low in this scenario. To influence tackle behaviour, it may also be necessary to limit the actions of the ball-carrier (eg, not allow them to lower their body height) to give the tackler an opportunity to target the torso, and in this case, it may also be worth considering whether very low tackles should be outlawed to protect the tackler from concussion.

Data for ball-carriers is consistent with previous evidence that 'head to head' contact and first point of contact being to the head and neck carry high odds of concussion. These findings suggest that strategies to lower tackle height are of greater benefit to the ball carrier than the tackler. This has previously been shown for all injuries, with lower tackles more likely to result in injury to the tackler and higher tackles more likely to result in injury to the ball-carrier.[6] However, the interaction of an upright tackler making contact with the head and neck of the ball-carrier was associated with risk of concussion to the tackler. Although the greatest benefit is likely to be when carrying the ball, this interaction shows that there is also likely to be a protective effect when tackling. This is consistent with current efforts to lower tackle height in elite rugby.

A whiplash mechanism was identified as having a high odds of ball-carrier concussion, and this has previously been highlighted as a relatively common mechanism of injury in women's university level rugby union.[26] This is the concussion mechanism that is most likely to be influenced by neck strengthening interventions and, while this might reduce the odds of a concussion, the fact that only 1.3% of ball-carrier concussions were linked to this mechanism should be considered.

Interaction analysis was limited by relatively small sample size once multiple characteristics were included; however, tackler speed was an important factor in tackler concussion unless the carrier was also travelling at high speed. There were 12 concussions, all to the tackler (5.6% of all tackler concussions), when both players were moving at high speed, with 6 of these tackles being from the side, 4 on an angle and 2 from the back. Approaching the ball-carrier at an angle, even when the ball-carrier is moving at high speed, is likely to reduce the chances of tackles sustaining a concussion compared with front-on tackles.[4] In contrast, when the ball-carrier was static, there was a higher odds of concussion to a high-speed tackler. There were only four events with this interaction of variables, and in all cases, the tackler moved more quickly than the rest of his defensive line to put extreme pressure on a ball-carrier. In all cases, the ball-carrier saw the tackler approaching, stopped moving and passed the ball to a teammate before the tackler arrived at such high speed that the tackle was uncontrolled and the tackle technique poor.

Limitations

Despite this analysis being carried out using broadcast quality video footage with multiple angles available, it was only possible to identify 54% of tackle concussion events on video. Assuming that concussion reporting was not skewed towards tackler or ball-carrier, there was a specific challenge in identifying ball-carrier concussive events, introducing bias towards the tackler when describing overall (tackler and ball-carrier) tackle characteristics with a high odds of concussion. At other levels of rugby, the quality of footage and number of camera angles can be a limitation,[27] but it is important to understand the risk of concussion in the tackle at other levels of the game given the different match and tackle characteristics.[28] A further limitation is that, despite having 5 years of league-wide data available, the sample was still relatively small, limiting our ability to explore interactions. The sample size required to detect interactions between two binary variables has been suggested to be fourfold that of a main effect of the same magnitude.[29] Therefore, very large samples are required to fully explore the dynamic tackle situation.

Policy implications

The main and novel contribution that this study makes to the literature is the exploration of tackles characteristics with high odds of resulting in concussion to tacklers and ball-carriers separately. There are ongoing policy discussions regarding strategies to reduce concussions in the tackle in both elite and community rugby union. This study shows that the odds of a concussion to the ball-carrier are lower when the point of contact on the ball-carrier is lower, providing strong support for a lowering of tackle height in elite rugby union. The odds of concussion to the tackler are greatest when their head contacts the hip or knee of the ball-carrier (ie, in lower tackles), whereas when contact is made with the torso, the odds of concussion are relatively low. This is consistent with the optimal target being the torso; however, targeting the torso rather than the shoulder region offers less margin for error for the tackler in avoiding head contact with the hip and knee and with a teammate's head when there is more than one tackler. The challenge is increased given dynamic changes in circumstances depending on the ball-carrier's behaviour. A further consideration is that players are most likely to change behaviour for performance benefit and therefore research that explores the association between tackle characteristics and performance outcomes will be an important contribution alongside the evidence regarding tackle characteristics and injury. For a lowering of tackle height to be effective in reducing concussions in tacklers, it is likely that it will also be necessary to limit the actions of the ball-carrier (eg, not allow them to lower their body height) as they enter the tackle to give the tackler an opportunity to target the torso.

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Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Ethics approval This study involves human participants and was approved by the Research Ethics Approval Committee for Health at the University of Bath approved the study (Ref: EP20-21 046). Participants gave informed consent to participate in the study before taking part.

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Data availability statement Data are available upon reasonable request.

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Author note Equity, diversity, and inclusion statement: the author group comprises eight men from early-career to experienced researchers drawn from sports medicine, sports science, epidemiology and performance analysis in academic and practice environments.

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